Arabidopsis Growth Under Nutrients and Damage

STAT 642 — Spring 2026
Team 11

Outline

1. Introduction & Data Description
2. Model
3. Analysis & Results
4. Assumption Checks
5. Conclusion

1.1 Introduction & Data Description

Source: Our source dataset is called `Arabidopsis`, and was generated from a **DoE experiment on Arabidopsis thaliana plants** performed by **the Center for Genomics and Systems Biology at New York University**, and the results of which were published in **Banta JA, Stevens MHH, Pigliucci M (2010). "A comprehensive test of the 'limiting resources' framework applied to plant tolerance to apical meristem damage." Oikos, 119(2), 359-369. doi:10.1111/j.16000706.2009.17726.x.**

Banta et al investigated how different resources **(nutrients and light)** affect the ability of Arabidopsis plants from three different regions to tolerate damage to their **apical meristem (AMD)** in order to test the "limiting resources" framework. The "limiting resources" framework was an accepted model in the field, describing the growth response of plants to AMD. The dataset contains measurements of plant growth and reproduction under various treatment conditions.

Sample size: $n = 625$

Unit of observation: individual plants

1.2 What Is Interesting About These Data?

- Plant response to **apical meristem damage (AMD)** reflects growth, survival, and adaptation
- AMD occurs due to **herbivory and environmental stress**
- Prior research shows **conflicting results**:
  - Some: higher nutrients $\rightarrow$ higher tolerance
  - Others: higher nutrients $\rightarrow$ lower tolerance
- This dataset allows a **controlled test** of how nutrients affect plant response to damage

Research Questions

Our research questions in analyzing this dataset are as follows:

1. How do the reproductive responses of Arabidopsis thaliana to apical meristem damage (AMD) vary across different regions, nutrient levels, and transplant sources?
2. Are there significant interactions between region, nutrient levels, and transplant sources in determining the plant's response to AMD?
3. How much of the variability in fruit production can be attributed to differences between populations and maternal seed families within regions?

1.3 Variables

Variable	Type	Description
Total Fruits	Response (integer)	The total number of fruits produced by the plant
Region	Fixed predictor	The region from which the plant was collected (NL, SP, SW)
Nutrients	Fixed predictor	The level of nutrients in the soil (1 prill = low, 8 prills = high)
AMD	Fixed predictor	Whether the plant was clipped to simulate apical meristem damage (clipped or unclipped)
Status	Fixed predictor	The source of the plant (Normal, Petri.Plate, or Transplant)
Rack	Fixed predictor	The light rack on which the plant was grown (1 or 2)
Population	Random grouping	The population of the plant, nested within Region (9 populations)
Genotype	Random grouping	The maternal seed family, nested within Population (24 families)

1.4 Experimental Design

Study organism: *Arabidopsis thaliana*

Sampling hierarchy

- 3 regions: NL, SP, SW
- 9 populations nested in Region (4 SP, 2 NL, 3 SW)
- 24 maternal seed families nested in Population

Treatment factors

- Nutrients: 1 prill (low) vs 8 prills (high)
- AMD: clipped vs unclipped
- Region: NL, SP, SW (fixed)

Nuisance factors (fixed)

- Rack: 2 light racks
- Status: Normal / Petri.Plate / Transplant

Random factors

- Population (9 levels)
- Genotype / maternal seed family (24 levels)

Design structure

A 2×2 factorial treatment (Nutrients $\times$ AMD) crossed with a 3-level sampling hierarchy. Region is fixed; Population and Genotype are random.

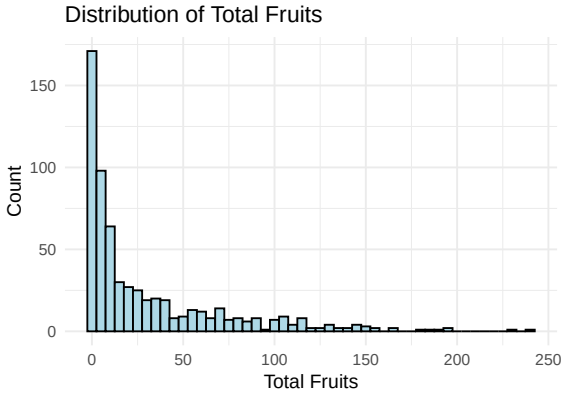
Procedure

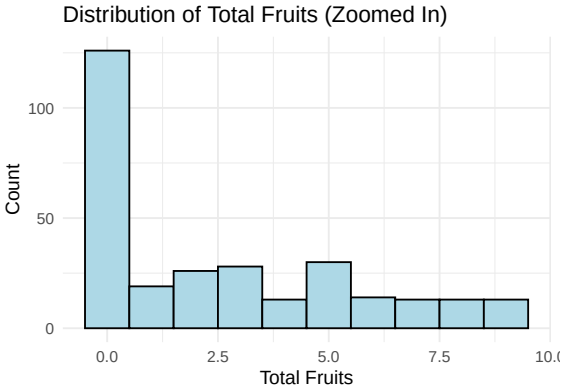
- Seeds germinated in petri dishes, then potted
- Failed germinations replaced with same-family seedlings
- Clipping applied at bolting stage

Scale: (n = 625); 6.5 plants per seed-family $\times$ treatment cell

1.5 Data Exploration & Results

Before fitting the model, we perform exploratory data analysis to understand the distribution of the response variable and the relationships between predictors and the response.





2.1 Model

Why a Mixed-Effects Model?

Fixed and random effects are combined to reflect the nesting: plants within **genotypes**, genotypes within **populations**, populations within **regions**.

Response: $\log(\text{Total Fruits} + 1)$; fit via `lme4::lmer` with Satterthwaite F-tests (`lmerTest`).

$$
\begin{aligned}
Y_{ijklmno} = \mu &+ \alpha_i + \beta_j + \gamma_k + \delta_l + \phi_m \\
&+ (\alpha\beta)_{ij} + (\alpha\phi)_{im} + (\beta\phi)_{jm} + (\alpha\beta\phi)_{ijm} \\
&+ (\alpha\gamma)_{ik} + (\beta\gamma)_{jk} + (\phi\gamma)_{mk} \\
&+ P_{n(i)} + F_{o(ni)} + \varepsilon_{ijklmno}
\end{aligned}
$$

Term definitions appear on the next slide.

2.2 Model Setup

Fixed main effects

- μ: overall intercept
- α_i: Region ($i = 1, 2, 3$; NL, SP, SW)
- β_j: Nutrients ($j = 1, 2$; low, high)
- ϕ_m: AMD ($m = 1, 2$; clipped, unclipped)
- γ_k: Status ($k = 1, 2, 3$)
- δ_l: Rack ($l = 1, 2$)

Random effects

- $P_{n(i)} \sim N(0, \sigma_P^2)$: Population in Region
- $F_{o(ni)} \sim N(0, \sigma_F^2)$: Seed family in Population
- $\varepsilon_{ijklmno} \sim N(0, \sigma^2)$: residual error

Treatment interactions

- $(\alpha\beta)_{ij}$: Region $\times$ Nutrients
- $(\alpha\phi)_{im}$: Region $\times$ AMD
- $(\beta\phi)_{jm}$: Nutrients $\times$ AMD
- $(\alpha\beta\phi)_{ijm}$: Region $\times$ Nutrients $\times$ AMD

Nuisance interactions

- $(\alpha\gamma)_{ik}$: Region $\times$ Status
- $(\beta\gamma)_{jk}$: Nutrients $\times$ Status
- $(\phi\gamma)_{mk}$: AMD $\times$ Status

2.3 Assumptions

As this data will be analyzed as a linear mixed-effects model, the following assumptions must be met:

1. **Linearity** — the relationship between predictors and each (transformed) response is linear
2. **Independence** — observations are independent conditional on the random effects
3. **Normality** — random effects and residual errors are normally distributed
4. **Equal Variance** — constant residual variance (homoscedasticity)
5. **Correct Model Specification** — the model includes all relevant predictors and interactions, and the random effects structure appropriately captures the nesting and variability in the data

The following additional assumptions are made due to the experimental design:

6. **Random Sampling within Strata** — within each region, populations and maternal seed families are assumed to be representative samples from the broader distribution of populations and families in that region, such that the random effects $P_{n(i)}$ and $F_{o(ni)}$ are exchangeable draws from their respective normal distributions
7. **Exogeneity of Random Effects** — the random effects are uncorrelated with the fixed effects

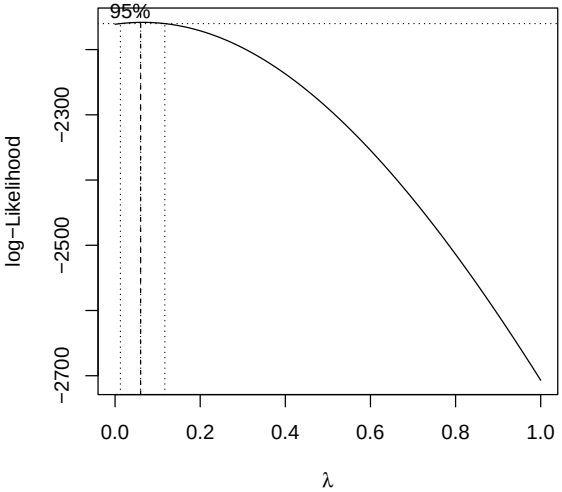
2.4 Model Fitting

Overview - Fit full LMM including: - Interactions of interest: Region $\times$ Nutrients, Region $\times$ AMD, Nutrients $\times$ AMD, Region $\times$ Nutrients $\times$ AMD - Nuisance interactions: Region $\times$ Status, Nutrients $\times$ Status, AMD $\times$ Status

- Use backward elimination (LRT, ($= 0.05$)) $\rightarrow$ obtain reduced model
- Model fitted with:
  - `lme4::lmer()`
  - `lmerTest` (Satterthwaite F-tests)

Response Transformation

- Use Box-Cox to assess appropriate transformation
- Result: $\lambda \approx 0.06 \approx 0 \rightarrow$ log transform
- Use $\log(\text{Total Fruits} + 1)$ to handle zeros



3.1 Analysis

Type III ANOVA

Table 2: Type III ANOVA for Full Model

	Sum Sq	Mean Sq	NumDF	DenDF	F value	Pr(>F)
reg	8.144	4.072	2	6.432	2.776	0.135
nutrient	166.774	166.774	1	592.870	113.714	0.000
amd	10.942	10.942	1	591.472	7.461	0.006
status	8.338	4.169	2	597.162	2.842	0.059
rack	84.520	84.520	1	589.588	57.630	0.000
reg:nutrient	14.052	7.026	2	591.062	4.791	0.009
reg:amd	0.639	0.320	2	588.625	0.218	0.804
nutrient:amd	5.431	5.431	1	589.090	3.703	0.055
reg:status	13.698	3.424	4	596.891	2.335	0.054
nutrient:status	1.262	0.631	2	592.691	0.430	0.651
amd:status	1.424	0.712	2	592.112	0.485	0.616
reg:nutrient:amd	3.579	1.789	2	588.685	1.220	0.296

3.2 Model Reduction

Backward elimination via LRT ($\alpha = 0.05$) on ML-fit models. Five interactions are tested and dropped; Region $\times$ Nutrients and Region $\times$ Status are retained.

Backward elimination (LRT)

Step	Interaction dropped	Chisq	Df	p
1	Nutrients $\times$ Status	0.89	2	0.640
2	AMD $\times$ Status	0.96	2	0.618
3	Region $\times$ Nutrients $\times$ AMD	2.69	2	0.261
4	Region $\times$ AMD	0.42	2	0.812
5	Nutrients $\times$ AMD	2.44	1	0.118

Reduced model: Type III tests

Effect	df1	df2	F	p
reg	2	6.4	2.78	0.135000
nutrient	1	600.7	148.97	0.000000
amd	1	597.2	11.10	0.000916
status	2	606.1	2.66	0.070700
rack	1	598.9	57.18	0.000000
reg:nutrient	2	599.9	4.77	0.008820
reg:status	4	605.9	2.51	0.040600

3.3 Model Reduction via AIC / BIC

- Enumerate all subsets of 7 candidate interactions
- **Hierarchy enforced**: if 3-way is in, all three 2-way interactions must also be in
- Main effects always retained; Rack additive
- 72 models (9 treatment $\times$ 8 nuisance); fit with ML
- AIC-preferred model adds **Nutrients $\times$ AMD** vs. the LRT-reduced model; not significant by LRT
- Retain LRT-reduced model for interpretability

Table 5: Top 10 of 72 models, ranked by AIC

Model	AIC	BIC	dAIC	dBIC
Reg x Nut + Nut x AMD + Reg x Stat	2058.91	2138.79	0.00	24.19
Reg x Nut + Reg x Stat	2059.35	2134.79	0.44	20.19
Reg x Nut	2060.92	2118.61	2.01	4.01
Reg x Nut + Nut x AMD	2060.94	2123.07	2.04	8.47
Reg x Nut + Nut x AMD + Reg x Stat + AMD x Stat	2061.84	2150.60	2.93	36.00
Reg x Nut + Nut x AMD + Reg x Stat + Nut x Stat	2062.03	2150.79	3.12	36.19
Reg x Nut + Reg x Stat + AMD x Stat	2062.26	2146.58	3.35	31.98
Reg x Nut + Reg x Stat + Nut x Stat	2062.44	2146.76	3.54	32.16
Reg x Nut + Reg x AMD + Nut x AMD + Reg x Stat	2062.49	2151.25	3.58	36.65
Reg x Nut + Reg x AMD + Reg x Stat	2062.93	2147.25	4.02	32.65

3.4 Alternative GLMM with Negative Binomial Family

- Total Fruits is a zero-heavy count $\rightarrow$ fit a **negative binomial GLMM** on the original scale
- Same fixed/random effects as the reduced LMM
- Goal: check whether a count model changes conclusions
- **Result:** similar conclusions for key fixed effects; LMM retained for interpretability

Table 6: Type III tests: NB GLMM

Effect	Chisq	Df	p-value
(Intercept)	47.37	1	<0.0001
reg	6.60	2	0.0368
nutrient	24.05	1	<0.0001
status	3.91	2	0.1416
amd	8.12	1	0.0044
rack	80.31	1	<0.0001
reg:nutrient	0.63	2	0.7287
reg:status	7.08	4	0.1318

3.5 Variance Components

Research question 3 asks how much variation in plant response is attributable to differences between populations and maternal seed families within regions. We answer it by partitioning the random-effect variance from the reduced LMM and testing each component via LRT.

Table 7: Variance components (reduced LMM)

Component	Variance	% of total
gen	0.024	1.3
popu	0.289	16.3
Residual	1.463	82.4

Table 8: LRT for random effects

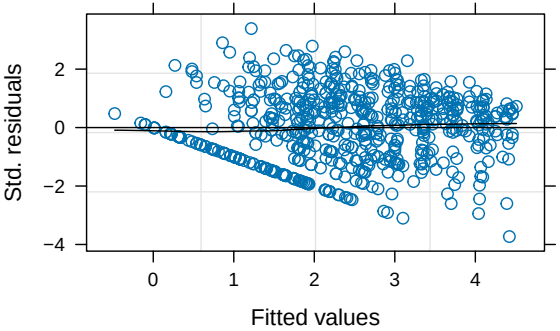
Term	LRT	Df	p-value
	NA	NA	NA
(1 | popu)	12.97	1	<0.001
(1 | gen)	1.48	1	0.224

Based on the variance components and LRT results, we attribute approximately 16% of the variance in fruit production to differences between populations. The LRT for the variance contributed by maternal seed families is not significant, suggesting that this level of grouping does not explain a significant portion of the variability in the response.

4.1 Assumption Checks

Residual Plots

- No strong nonlinear pattern
- Pattern reflects zero-inflation (not model misspecification)
- Residuals approximately normal
- Slight variance increase at higher fitted values



4.2 Variance Checks

- Brown–Forsythe (Levene) test:
  - Some heteroscedasticity across Region and Status
  - Variance ratios $< 1.5 \rightarrow$ not severe
- Large sample size (n = 625):
  - Test is sensitive to small differences*

Final Assessment

- Normality: approximately satisfied
- Independence: satisfied by design
- Equal variance: mild violations

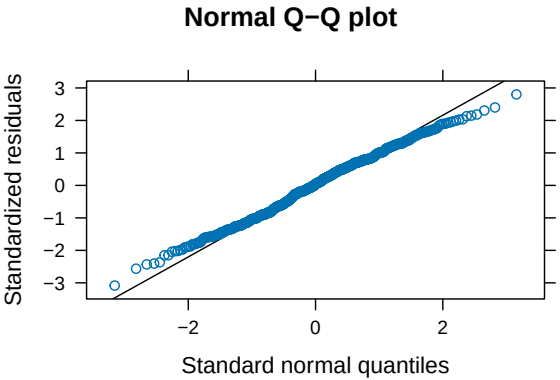
$\rightarrow$ Overall, model assumptions are **reasonably satisfied**

4.3 Assumption Checks: Q–Q plot

- Q–Q plot shows residuals $\approx$ normal
- Shapiro–Wilk test ($p = 0.03$) attributed to large sample size (n=625) and zero-inflation, not severe non-normality

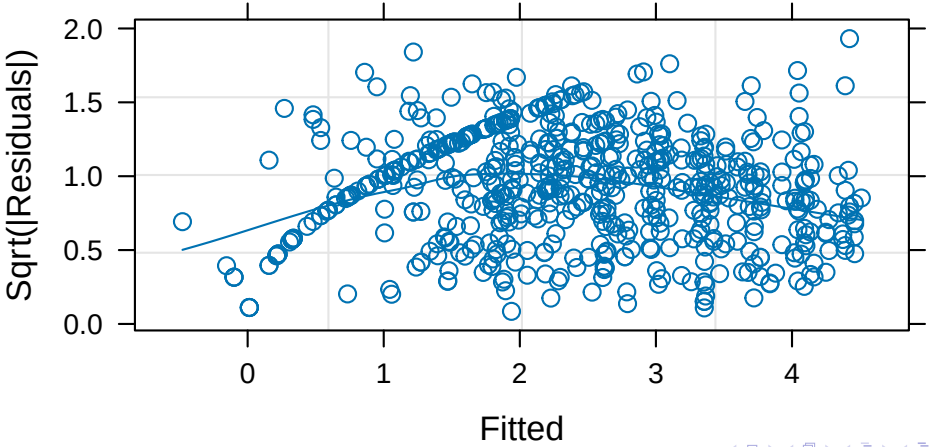
```
    Shapiro-Wilk normality test

data:  resids
W = 0.99466, p-value = 0.02797
```



4.4 Assumption Checks: Variance

- Scale-location plot:
  - Slight increase in variance at moderate fitted values
  - No severe heteroscedasticity pattern



4.5 Conclusion

Levene Test

Factor	p_value
Region	0.0059
Nutrients	0.7100
Status	0.0245
AMD	0.3870

Final Assessment

Group	Variance_Ratio
Region	1.38
Status	1.43

Conclusion

- Log transformation addresses skewness and zeros
- Mild heteroscedasticity, but not severe
- GLMM confirms robustness of results

→ Model assumptions reasonably satisfied

4.6 Hypothesis Tests — Type III F-tests

Table 11: Type III F-tests for log(Total Fruits + 1)

Effect	F	df1	df2	p-value
reg	2.781	2	6.41	0.135
nutrient	148.973	1	600.74	<0.001
amd	11.102	1	597.19	<0.001
status	2.661	2	606.10	0.071
rack	57.176	1	598.90	<0.001
reg:nutrient	4.768	2	599.90	0.009
reg:status	2.514	4	605.95	0.041

Significant Effects

- Significant: **Nutrients, AMD, Rack, and Status**
- Significant interactions:
  - Region × Nutrients
  - Region × Status

4.7 Post-hoc Analysis (EMMs)

For significant effects, examine **estimated marginal means** on the log(Total Fruits + 1) scale.

Nutrients within Region

reg	contrast	estimate	SE	p.value
NL	nutrient1 - nutrient8	-1.562	0.232	
SP	nutrient1 - nutrient8	-1.458	0.142	
SW	nutrient1 - nutrient8	-1.859	0.165	

AMD (main effect)

contrast	estimate	SE	p.value
clipped - unclipped	-0.324	0.097	0.001

Status within Region

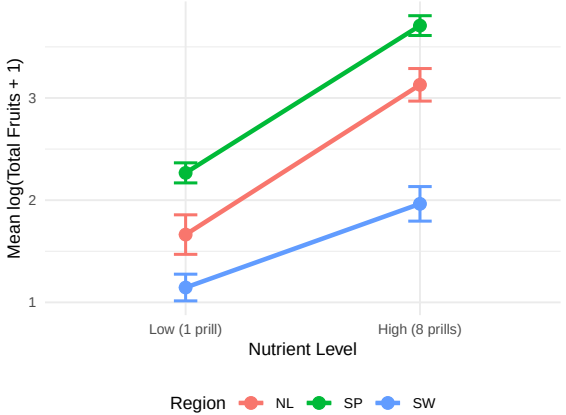
reg	contrast	estimate	SE	p
NL	Normal - Petri.Plate	-0.180	0.294	
NL	Normal - Transplant	0.402	0.294	
NL	Petri.Plate - Transplant	0.582	0.360	
SP	Normal - Petri.Plate	0.417	0.204	
SP	Normal - Transplant	0.370	0.182	
SP	Petri.Plate - Transplant	-0.047	0.244	
SW	Normal - Petri.Plate	0.636	0.287	
SW	Normal - Transplant	-0.113	0.188	
SW	Petri.Plate - Transplant	-0.749	0.312	

4.8 Interaction Plots

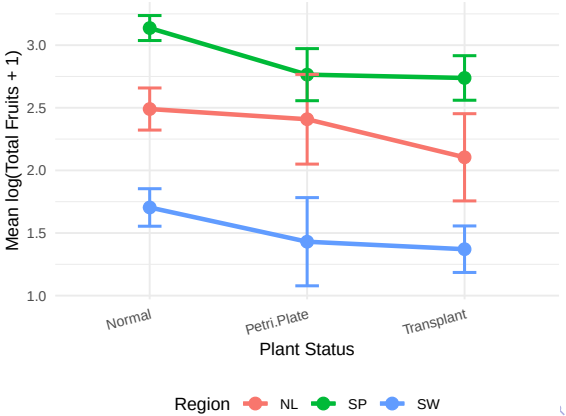
Visualizing Significant Interactions

The interaction plots below illustrate how the effects of **Nutrients** and **Status** on fruit production vary across **Regions**.

Region × Nutrients



Region x Status



5.1 Summary of Findings

- **Nutrients, AMD, Rack, and Status** were all found to have significant impacts on fruit production. Plants grown in the presence of high nutrients and in the absence of AMD produced significantly more fruits than those under low nutrients and exposed to AMD.
- **Region alone was not significant**, but its role became important through interactions with other predictors (Nutrients and Status).
- Overall, fruit production in *Arabidopsis* was shaped by both **experimental treatments** and **hierarchical biological differences** among regions and populations.

5.2 Research Question Conclusions

1. How do the reproductive responses of Arabidopsis thaliana to apical meristem damage (AMD) vary across different regions, nutrient levels, and transplant sources?
   - No significant interactions between AMD and Region, Nutrients, or Status were observed, suggesting the effect of AMD on fruit production was consistent across these factors.
2. Are there significant interactions between region, nutrient levels, and transplant sources in determining a plant's fruit production?
   - Yes, significant interactions were found between Region $\times$ Nutrients and Region $\times$ Status, indicating that the effects of nutrient availability and transplant source varied across regions.
3. How much of the variability in plant response to AMD can be attributed to differences between populations and maternal seed families within regions?
   - Approximately 16% of the variance in fruit production was attributable to differences between populations, while the variance attributed to maternal seed families was not significant.

5.3 Limitations & Future Work

- Residual diagnostics suggested **some heteroscedasticity across regions and status groups**, although the variance differences were not extreme.
- Because the response variable is a **count with many zeros**, transformation-based Gaussian modeling may not fully capture the data structure.
- While the linear mixed-effects model was interpretable and supported the main conclusions, future work should further investigate **generalized mixed models for count outcomes**.

5.4 References

- Banta, J. A., Stevens, M. H. H., & Pigliucci, M. (2010).
  A comprehensive test of the "limiting resources" framework applied to plant tolerance to apical meristem damage.
  Oikos, 119(2), 359–369.
- Dean, A., Voss, D., & Draguljić, D. (2017).
  Design and Analysis of Experiments (2nd ed.). Springer.
- STAT 642 Course Handouts, including:
  - Handout 6: Random Effects Models
  - Handout 9: Mixed Models and Variance Components
  - Handout 10: Model Construction and Inference